

Christelle Abou Rjeily Helou

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SUMMARY

Graduate in Artificial Intelligence and Physics, with proven and tested self taught skills in coding, research and data collection.

EDUCATION

MSc in Artificial Intelligence | University of West England Bristol

September 2023 - September 2024

Grade Classification: First-Class Honours Degree; Distinction

BSc in Physics | Notre-Dame Louaize University

Sep 2020 - May 2023

CGPA: 3.31/4; Honours: On the Dean's List

Vice-President of the NDU Astronomy Club

Lebanese Baccalaureate in General Sciences | Ecole des Soeurs de la Charité Besancon

Sep 2004 - Jun 2020

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL (MySQL), R

Data Science & Machine Learning: Machine Learning, Big Data Analytics, Natural Language Processing, Knowledge-Based and Hybrid Systems, Clustering Algorithms, Similarity Measures

Development Tools & Platforms: Visual Studio, Anaconda, RStudio, MATLAB, $\text{\LaTeX}$

Microsoft Office Suite: Word, Excel, PowerPoint

Libraries and Frameworks: PyTorch, Keras, pandas, scikit, NumPy, Sklearn, TensorFlow, Hugging Face Transformers, NLTK

Natural Language Processing: BERT, LSTM, Attention mechanisms, Feature extraction, Recursive Feature Elimination

Large Language Models: Fine-tuning, prompt engineering, transformer architectures

Fuzzy Systems: ANFIS, Fuzzy logic implementation

Familiar OS: Windows, Linux

PROJECTS

Cosmic Expansion Simulation [2025] A Python project that simulates the expansion of the Universe using the Friedmann equations under different cosmological models. (Ongoing)

Seismic Detection Across the Solar System. [2024] This project was conducted to identify seismic quakes within the noise in real data from the Apollo missions and the Mars InSight Lander using classical machine learning approaches such as convolutional neural network. This project was a part of the NASA SpaceApps Challenge 2024. (October 2024)

Text Classification with Transformers. [2024] Implemented and fine-tuned BERT models using Hugging Face Transformers library for multi-class emotion classification, achieving improved performance over traditional approaches through attention mechanisms and contextual embeddings.

Masters Dissertation: Sentiment analysis. [2024] This research was conducted for MSc Artificial Intelligence at University of West of England - Bristol. Developed a novel fuzzy-hybrid approach combin-

ing BERT transformers, LSTM networks, and Support Vector Machines with lexicon-based methods for emotion classification. Implemented ANFIS to handle uncertainty in emotional states using Plutchik's 8-emotion model. Applied RFE for optimal feature selection and compared performance against Bi-LSTM with attention mechanisms to classify complex emotional expressions beyond traditional sentiment analysis. (September 2024)

Music Recommendation using Hybrid Systems. [2024] This project was conducted under the Knowledge and Hybrid Systems module in MSc Artificial Intelligence at University of West of England - Bristol. The main objective behind it is to develop a comprehensive hybrid recommendation system combining K-means clustering, cosine similarity, and Fuzzy Inference Systems to provide personalized music recommendations. Implemented TF-IDF vectorisation for artist term processing, applied K-means clustering for genre segmentation, and utilised cosine similarity for feature-based song matching as well as creating a two-layer fuzzy logic system handling user preferences for tempo, loudness, duration, and decade preferences. (June 2024)

3D MoonQuake Map. [2022 - 2023] Aberration.dev is a constantly updated website that was developed for the purpose of visualizing the moonquakes caused by several different events. JavaScript and Python were used in order to code the simulation, more specifically Three.js and Obspy and many other libraries. This project was a part of the NASA SpaceApps Challenge 2022 and 2023. ([Try it here](#))

Bachelors Dissertation: Open Cluster NGC 2420. [2023] This research was conducted for the purpose of finding the age of the open cluster NGC 2420. Python and Glue software were used in order to fit isochrones to the colour magnitude diagram plotted from SQL files of NGC 2420 data. This project was a part of my senior to conclude my bachelor degree in physics.

JobRCT. [2023] JobRCT is a website that was developed for the purpose of simplifying writing job descriptions for hiring managers. PHP and HTML along with many other languages were used in order to design the website and develop it, along with SQL for the databases. This project was part of an internship for the company STIRA.

WORK EXPERIENCE

Home Bargains Oct 2023 - Present

Part-time Retail team member.
Locations: Bristol - Brislington and Petherbridge Way (Muller Road)
Scotland - Forge Glasgow

SIIRA Company Internship June 2023 - August 2023

Worked on a prompting and AI Project: JobGPT.

Airport Internship May 2023

Overview of the work done in the radar room and the tower with a full report describing it.

Pharmacist Assistant Sep 2020 - Jan 2021

Data Entry; dispensing drugs to the patients; controlling the inventory; supervising expiry dates.

ACTIVITIES

NASA SpaceApps Challenge Oct 2022 - Oct 2023

Topic : Make a MoonQuake Map (Team Abberation)
Years : 2022 and 2023

Achievements : Winners of the golden local award, honourable mentions globally (top 30s)

Vice President of Astronomy Club at Notre Dame Univesity Louaize 2021 - 2023

Organized and carried out astronomy activities such as stargazing, movie and quiz nights, observatory visits, and enriched the community with scientific astro knowledge)

Computer Science Summer Camp Jul 2019

Main Topic : Python

Model United Nations Mar 2019

Topic: Agriculture - Egypt

Institute of Citizenship ADYAN 2017 - 2019

Engaged in a non-formal education program that aims to foster the concept of inclusive and active citizenship; worked on expanding knowledge regarding diversity and accepting others.

Volunteer | SOS children's village Lebanon Dec 2017

Responsibilities : collected and organized the donations and took care of the children.

School Social Program 2015 - 2020

Responsibilities : worked with donations and unfortunate men and women, elders and children.

Achievements : tutored an exchange student with disabilities in order to pass class 9.

Mixed Martial Arts and Brazilian Jiu Jitsu | Team Braga Lebanon

SCHOLARSHIPS AND AWARDS

Academic Scholarship at University of the West of England Bristol 2023 - 2024

NASA Space Apps Challenge Global Finalists Honorable Mentions 2023

Volunteering Certification by Caritas Lebanon 2023

Notre Dame University Honoring Ceremony for success in NASA Space Apps 2023

NASA Space Apps Challenge Global Finalists Honorable Mentions 2022

Academic Scholarship at Notre Dame University - Louaize 2020 - 2023

Model United Nations 2019

LANGUAGES

English: Proficient User

IELTS Scores:

Overall Band Score: 8.5

Listening: 9.0

Reading: 9.0

Writing: 7.0

Speaking: 8.0

French: Fluent

Arabic: Proficient User

German: B1 Level